1 Basic matrix operations

1.1 Addition

$$\underset{n\times m}{\mathbf{A}} + \underset{n\times m}{\mathbf{B}} = \underset{n\times m}{\mathbf{C}} \tag{1.1}$$

In terms of elements

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}}_{\mathbf{A}} + \underbrace{\begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \end{bmatrix}}_{\mathbf{B}} = \underbrace{\begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \cdots & a_{1m}+b_{1m} \\ a_{21}+b_{21} & a_{22}+b_{22} & \cdots & a_{2m}+b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}+b_{n1} & a_{n2}+b_{n2} & \cdots & a_{nm}+b_{nm} \end{bmatrix}}_{\mathbf{C}}$$

where row i and column j of $\mathbf{C}$ is the element

$$c_{ij} = a_{ij} + b_{ij}$$

1.2 Multiplication with scalar

For scalar c and $n \times 1$ vector $\boldsymbol{x}$

$$c \underset{n\times 1}{\boldsymbol{x}} = \begin{bmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{bmatrix} \tag{1.2}$$

For scalar c and $n \times m$ matrix $\mathbf{A}$

$$c \underset{n\times m}{\mathbf{A}} = \underbrace{\begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1m} \\ ca_{21} & ca_{22} & \cdots & ca_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{n1} & ca_{n2} & \cdots & ca_{nm} \end{bmatrix}}_{c\mathbf{A}} \tag{1.3}$$

The matrix $\mathbf{C} = c\mathbf{A} + k\mathbf{B}$ In terms of elements

$$\underbrace{\begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1m} \\ ca_{21} & ca_{22} & \cdots & ca_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{n1} & ca_{n2} & \cdots & ca_{nm} \end{bmatrix}}_{c\mathbf{A}} + \underbrace{\begin{bmatrix} kb_{11} & kb_{12} & \cdots & kb_{1m} \\ kb_{21} & kb_{22} & \cdots & kb_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ kb_{n1} & kb_{n2} & \cdots & kb_{nm} \end{bmatrix}}_{k\mathbf{B}} = \underbrace{\begin{bmatrix} ca_{11}+kb_{11} & ca_{12}+kb_{12} & \cdots & ca_{1m}+kb_{1m} \\ ca_{21}+kb_{21} & ca_{22}+kb_{22} & \cdots & ca_{2m}+kb_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{n1}+kb_{n1} & ca_{n2}+kb_{n2} & \cdots & ca_{nm}+kb_{nm} \end{bmatrix}}_{\mathbf{C}}$$

1.3 Transpose

For $n \times m$ matrix $\mathbf{A}$,

$$\mathbf{A} = \begin{bmatrix} — & \boldsymbol{a}_1 & — \\ — & \boldsymbol{a}_2 & — \\ & \vdots & \\ — & \boldsymbol{a}_n & — \end{bmatrix}_{n\times m}, \ \mathbf{A}^\top = \begin{bmatrix} | & | & & | \\ \boldsymbol{a}_1^\top & \boldsymbol{a}_2^\top & \cdots & \boldsymbol{a}_n^\top \\ | & | & & | \end{bmatrix}_{m\times n}$$

where $\mathbf{A}^\top$ is *the transpose of* $\mathbf{A}$. In elements

$$\mathbf{A}^\top = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}^\top = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1m} & a_{2m} & \cdots & a_{nm} \end{bmatrix}_{m\times n}$$

1.4 Addition of matrices in terms of vectors

Write

$$\mathbf{A} = \begin{bmatrix} | & | & & | \\ \boldsymbol{a}_1 & \boldsymbol{a}_2 & \cdots & \boldsymbol{a}_m \\ | & | & & | \end{bmatrix}_{n\times m}, \text{ and } \mathbf{B} = \begin{bmatrix} | & | & & | \\ \boldsymbol{b}_1 & \boldsymbol{b}_2 & \cdots & \boldsymbol{b}_m \\ | & | & & | \end{bmatrix}_{n\times m}$$

then

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} | & & | & | & & | & & | & & | \\ \boldsymbol{a}_1 & + & \boldsymbol{b}_1 & \boldsymbol{a}_2 & + & \boldsymbol{b}_2 & \cdots & \boldsymbol{a}_m & + & \boldsymbol{b}_m \\ | & & | & | & & | & & | & & | \end{bmatrix}_{n\times m}$$

1.5 Inner product of vectors

For $n \times 1$ vectors $\boldsymbol{x}$ and $\boldsymbol{y}$

$$\boldsymbol{x}^\top \boldsymbol{y} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^{n} x_i y_i \tag{1.4}$$

1.6 Outer product of vectors

For $n \times 1$ vectors $\boldsymbol{x}$ and $\boldsymbol{y}$

$$\boldsymbol{x}\boldsymbol{y}^\top = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix} = \begin{bmatrix} x_1y_1 & x_1y_2 & \cdots & x_1y_n \\ x_2y_1 & x_2y_2 & \cdots & x_2y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_ny_1 & x_ny_2 & \cdots & x_ny_n \end{bmatrix}_{n\times n} \tag{1.5}$$

where element (i, j) is $(\boldsymbol{x}\boldsymbol{y}^\top)_{ij} = x_iy_j$

1.7 Inner product of matrices

For $n \times p$ matrix $\mathbf{A}$ and $p \times m$ matrix $\mathbf{B}$, yields $n \times m$ matrix (*NB: inner dimension p needs to be the same*)

$$\underset{n\times p}{\mathbf{A}}\underset{p\times m}{\mathbf{B}} = \underset{n\times m}{\mathbf{C}} \tag{1.6}$$

with element c_{ij} in row i and column j of $\mathbf{C}$

$$c_{ij} = \sum_{k=1}^{p} a_{ik}b_{kj} \tag{1.7}$$

In terms of vectors

$$\mathbf{A} = \begin{bmatrix} — & \boldsymbol{a}_1 & — \\ — & \boldsymbol{a}_2 & — \\ & \vdots & \\ — & \boldsymbol{a}_n & — \end{bmatrix}_{n\times p}, \mathbf{B} = \begin{bmatrix} | & | & & | \\ \boldsymbol{b}_1 & \boldsymbol{b}_2 & \cdots & \boldsymbol{b}_n \\ | & | & & | \end{bmatrix}_{p\times m}$$

$$\underset{n\times p}{\mathbf{A}}\underset{p\times m}{\mathbf{B}} = \begin{bmatrix} \boldsymbol{a}_1\boldsymbol{b}_1 & \boldsymbol{a}_1\boldsymbol{b}_2 & \cdots & \boldsymbol{a}_1\boldsymbol{b}_m \\ \boldsymbol{a}_2\boldsymbol{b}_1 & \boldsymbol{a}_2\boldsymbol{b}_2 & \cdots & \boldsymbol{a}_2\boldsymbol{b}_m \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{a}_n\boldsymbol{b}_1 & \boldsymbol{a}_n\boldsymbol{b}_2 & \cdots & \boldsymbol{a}_n\boldsymbol{b}_m \end{bmatrix}_{n\times m}$$

Rule of transpose:

$$(\mathbf{A}\mathbf{B})^\top = \mathbf{B}^\top\mathbf{A}^\top \tag{1.8}$$

Combining the definition of multiplication and addition we get the distributive Law

$$\underset{n\times p}{\mathbf{A}}(\underset{p\times m}{\mathbf{B}} + \underset{p\times m}{\mathbf{C}}) = \mathbf{AB} + \mathbf{AC}$$

and,

$$(\mathbf{B} + \mathbf{C})\mathbf{A}^\top = \mathbf{BA}^\top + \mathbf{CA}^\top$$